

Practical No: 03

Aim:

To retrieve documents from a MongoDB collection using basic query operations.

Theory

Querying is one of the most essential operations in any database system, as it allows users to retrieve meaningful information from stored data. In MongoDB, the primary method used to retrieve documents from a collection is the **find()** function. Unlike relational databases that use SQL, MongoDB uses a query language based on JSON-like syntax. This makes querying intuitive, flexible, and consistent with the document structure in which data is stored.

The **find()** method returns documents from a collection that match the specified query filter. When called without any parameters, `db.collection.find()` returns **all documents** present in the collection. This is equivalent to selecting all rows in an SQL table using `SELECT *`. The output of `find()` is displayed in BSON/JSON format, making the data easy to read and understand. By default, the output is not formatted; however, MongoDB provides the `pretty()` method to make the results more readable.

MongoDB queries are built on the principle of **key-value matching**. A basic query consists of a field and value pair—for example, `{ dept: "CSE" }`. This query filters the documents and returns only those where the value of the `dept` field is "CSE". MongoDB performs this search efficiently using indexes where available, or through document scanning when no index is defined. The dynamic schema of MongoDB allows documents to contain varying fields, and the query engine handles this flexibility without requiring fixed structures or strict typing.

Commands:

```
db.students.find()
```

```
db.students.find( { dept: "CSE" } )
```

Expected Output:

JSON documents matching the filter.

Viva Questions & Answers

Q1. What command is used to retrieve documents in MongoDB?

The `find()` command is used to retrieve documents from a MongoDB collection.

Q2. What does `db.collection.find()` do without any parameters?

It retrieves and displays all documents present in the collection.

Q3. How do you retrieve documents with a specific condition?

By passing a filter object inside `find()`, for example:

```
db.students.find( { dept: "CSE" } )
```

Q4. What is the purpose of the pretty() method?

The pretty() method formats the output in a readable JSON structure.

Q5. How are MongoDB queries different from SQL queries?

MongoDB uses JSON-like syntax for queries instead of SQL statements.

Conclusion:

In this practical, we successfully retrieved documents from a MongoDB collection using basic query operations. By using the find() method, we were able to display all documents as well as filter specific records based on given conditions.